short fibres coming out of the cell body are called **dendrites,** which transmit impulses towards the cell body. The cell body and the dendrites contain cytoplasm and granulated endoplasmic reticulum called **Nissl's granules.**

An axon is a long fibre that arises from a cone shaped area of the cell body called the **Axon hillock** and ends at the branched distal end. Axon hillock is the place where the nerve impulse is generated in the motor neurons. The axon of one-neuron branches and forms connections with many other neurons. An axon contains the same organelles found in the dendrites and cell body but lacks Nissl's granules and Golgi apparatus.

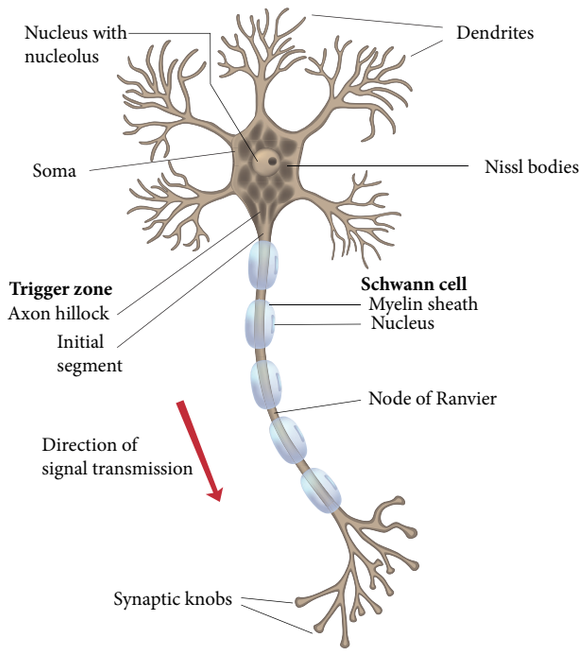


Figure 10.1 Neuron

The axon, particularly of peripheral nerves is surrounded by **Schwann cells** (a type of glial cell) to form myelin sheath, which act as an insulator. **Myelin sheath** is associated only with the axon; dendrites are always non-myelinated. Schwann cells are not continuous along the axon; so there are gaps in the myelin sheath between adjacent Schwann cells. These gaps are called **Nodes of Ranvier**. Large myelinated nerve fibres conduct impulses rapidly, whereas non-myelinated fibres conduct impulses quite slowly (Figure 10.1).

The longest cells in the human body are the **neurons**. The **longest** axons in the **human body**, for example, are those of the sciatic nerve, which run from the base of the spine to the big toe of each foot. These single-cell fibers may extend a meter or even longer. The axons of the inter neurons in the CNS are the shortest.

Each branch at the distal end of the axon terminates into a bulb like structure called **synaptic knob** which possesses **synaptic vesicles** filled with **neurotransmitters**. The axon transmits nerve impulses away from the cell body to an **inter neural space** or to a **neuro-muscular junction.**

The neurons are divided into three types based on number of axon and dendrites they possess (Figure.10.2).

1. Multipolar neurons have many processes with one axon and two or more dendrites. They are mostly interneurons.

2. Bipolar neurons have two processes with one axon and one dendrite. These are found in the retina of the eye, inner ear and the olfactory area of the brain.

3. Unipolar neurons have a single short process and one axon. Unipolar neurons are located in the ganglia of cranial and spinal nerves.

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